

Circular Motion

Angular Quantities

When dealing with circular motion, angular quantities are used to describe the motion.

For uniform circular motion, you deal with:

Radians: angle subtended at center
When arc length = radius

- deg $\rightarrow$ rad : $\times \frac{\pi}{180}$

**ALWAYS
USE radians
in formulas**

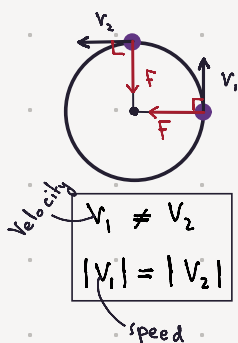
Uniform Circular Motion

- Speed (magnitude) is constant

- Velocity is NOT constant

→ Constant magnitude
→ changing direction

- **Condition:** Resultant force and acceleration are perpendicular to velocity

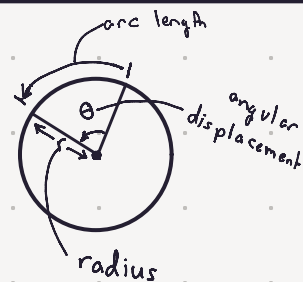


① Angular Displacement (θ)

- The angle subtended at center by arc 's'

- Formula: $\theta = \frac{s}{r}$

- Units: rad - 1 revolution: $\theta = 2\pi$



② Angular Speed (ω)

- Rate of change of angular displacement

- Formula: $\omega = \frac{\Delta\theta}{\Delta t} = \frac{v}{r} \rightarrow v = \omega r$

- Units: rad s⁻¹

- 1 revolution: $\omega = \frac{2\pi}{T}$ (Period for 1 rev)

Centripetal Acceleration (a_c)

- Defines Acceleration perpendicular to velocity

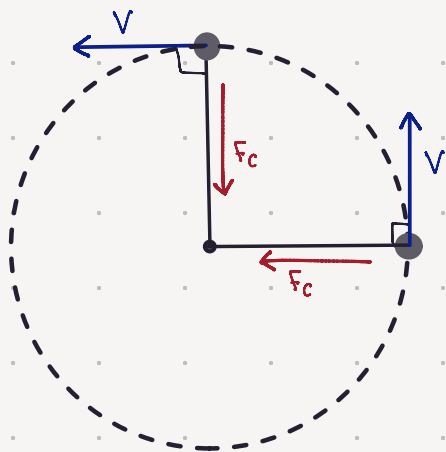
- Formula: $a_c = \frac{v^2}{r} = r\omega^2$

- Direction: Perpendicular to velocity (center)

- Affects change direction of velocity

Centripetal Acceleration & Force

As mentioned before, velocity is changing due to change in direction. Therefore there must be acceleration and a force that point perpendicular towards center. These are centripetal acceleration ① and centripetal force ②.



Centripetal Force (F_c)

- Formula: $F_c = ma_c = \frac{mv^2}{r} = mr\omega^2$

- Source: Tension, Friction, Normal force

□ Tension (string) □ Friction (Flat road)

□ Normal force (Inclined track) □ Gravity

- Direction: toward center

- Affects change direction of velocity

- Key Relationships:

□ $F \propto \frac{1}{r}$: $C \uparrow$ radius $\rightarrow$ less force needed

□ $F \propto v^2$: $C \times 2 \times v \rightarrow 4 \times$ force needed

□ $F \propto m$: $C \uparrow$ mass $\rightarrow$ more force needed

- What if (constant v):

□ $F_{\text{provided}} = F_{\text{required}} \rightarrow$ Object stays on track

□ $F_{\text{provided}} > F_{\text{required}} \rightarrow$ Object moves inwards

□ $F_{\text{provided}} < F_{\text{required}} \rightarrow$ Object moves outwards

□ $F_{\text{provided}} = 0 \rightarrow$ Object moves in a straight line

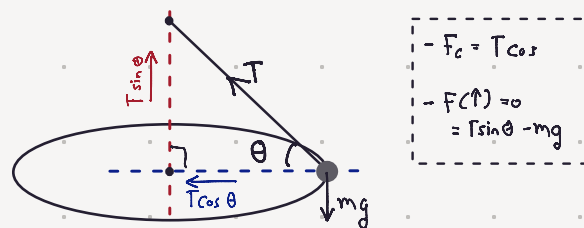
Solving Circular Motion Problems

Step 1: Identify All Forces

Some common forces:

- Weight, mg → Vertically Downwards
- Tension, T → Along string/rod
- Normal, N → Perpendicular to surface
- Friction, f → Opposes motion/tendency
- Gravitational → $\frac{GMm}{r^2}$
more details in Chapter 13

Step 2: Draw Axes

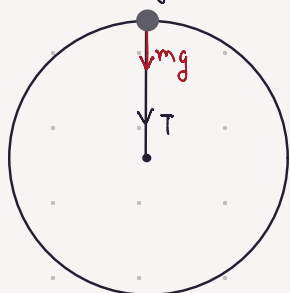


$$\begin{aligned} -F_c &= T \cos \theta \\ -F(\uparrow) &= 0 \\ &= T \sin \theta - mg \end{aligned}$$

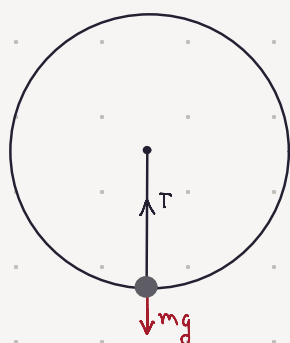
- ① Draw "main" axis through center and object
- ② Draw perpendicular axes
- ③ Resolve → Sum of forces along main axis = F_c

Vertical Circle [examples]

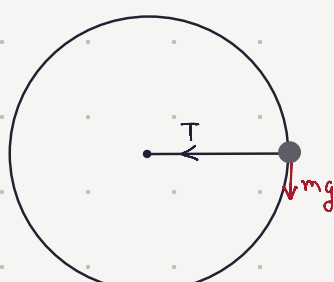
① $F_c = mg + T$



② $F_c = T - mg$

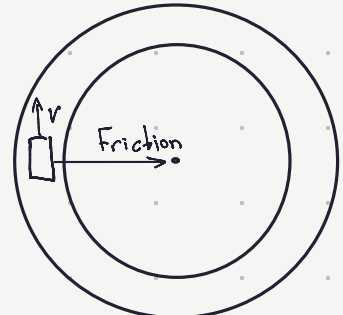


③ $F_c = T$

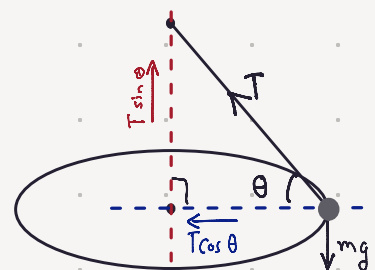


Horizontal Circle [examples]

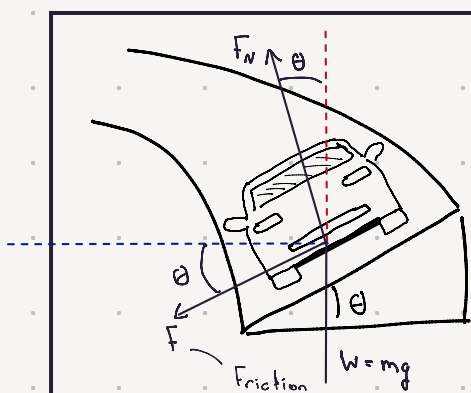
① Friction = F_c



② $F_c = T \cos \theta$

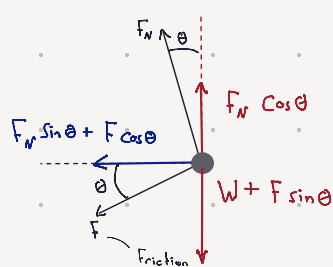


note that here $mg = T \sin \theta$
 so $F(\uparrow) = 0$



Forces: Friction, gravitational, Normal

Resolve:



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